MILITARY SPECIFICATION

STUFFING TUBES FOR USE WITH CIRCULAR ANODES AND REFERENCE ELECTRODES (CORROSION PREVENTIVE).

1. SCOPE

1.1 This specification covers the requirements for stuffing tubes to be used for attachment of circular platinum-palladium anodes and reference electrodes to a ship's hull or similar structure for corrosion protection (cathodic protection) of the hull or structure. It also includes features of design and fabrication of these stuffing tubes including alternate components.

1.2 Classification. - The stuffing tubes shall be furnished in the following types and classes, as specified (see 6.2):

Type 1 - New Installations.
Type 2 - Installations With Existing Gland.

Class A - Platinum-Palladium Anodes.
Class B - Silver-Silver Chloride Reference Electrodes.
Class C - Zinc Reference Electrodes.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

NN-P-71 - Pallets, Materials-Handling, Wood, General Construction Requirements.
QQ-N-281 - Nickel-Copper-Alloy (Monel and R-Monel) Bars, Plates, Rods, Sheets, Strips, Wire, Forgings, and Structural and Special Shaped Sections.
QQ-S-365 - Silver Plating, (Electrodeposited).
QQ-Z-325 - Zinc Coating, Electrodeposited, Requirements for.
PPP-B-601 - Boxes, Wood, Cleated-Plywood.
PPP-B-621 - Boxes, Wood, Nailed and Lock-corner.
PPP-B-636 - Box, Fiberboard.

MILITARY

JAN-P-100 - Packaging and Packing for Overseas Shipment, General Specification.
MIL-S-890 - Steel: Forgings and Bars for Hull, Engines, and Ordnance (Heat Treated).
MIL-C-915 - Cable, Cord and Wire, Electrical (Shipboard Use).
MIL-P-15011 - Pallets, Material-Handling, Hardwood, Post Construction, 4-way.
MIL-S-16974 - Steel Bars, Billets, Blooms and Slabs, Carbon and Alloy for Reforging or other Operations before Heat Treatment.
MIL-P-19468 - Plastic Rods, Polyetrafluoroethylene, Molded and Extruded.
MIL-S-20166 - Steel Bars and Shapes (for Hull Construction) (Including Material for Drop and Miscellaneous Forgings).

STANDARDS

MILITARY

MIL-STD-129 - Marking for Shipment and Storage

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

FSC 2040

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2.2 Other publications. - The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

OFFICIAL CLASSIFICATION COMMITTEE
Uniform Freight Classification Rules.

(Application for copies should be addressed to the Official Classification Committee, 1 Park Avenue at 33rd Street, New York 16, New York.).

NATIONAL BUREAU OF STANDARDS
Handbook H28 - Screw Thread Standards for Federal Services.

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington 25, D.C.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS
ASTM B-139-58 - Specification for Phosphor Bronze Rod, Bar and Shapes.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia 3, Pennsylvania.)

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

3. REQUIREMENTS

3.1 Description. - The stuffing tube shall consist of a steel body designed to receive the projecting hub and pin of a circular anode or reference electrode and to hold the end of the pin in contact with an electrical cable that will transmit current from or to other components of a cathodic protection system. The steel body shall contain a teflon bushing, connector assembly with compression ring, molded packing, and gland nut assembly with gland ring and spring lock washer, all arranged to accomplish the stated purpose. An electrical cable of required length shall be fitted to the tube. The stuffing tube dimensions and assembly shall conform to figure 1.

3.2 Materials. -

3.2.1 Body (all types). - The body shall be steel conforming to class HT of MIL-S-20166. An alternate material shall be steel conforming to grade 8620 of MIL-S-16974, that has been heat-treated to obtain the physical properties of steel conforming to class AN of MIL-S-890. On hulls where this material will not be compatible due to welding or similar effects, i.e., aluminum, an alternate material shall be as specified (see 6.2).

3.2.2 Bushing (all types). - The bushing shall be teflon (polytetrafluoroethylene) conforming to group B of MIL-P-19468.

3.2.3 Connector assembly. - Each connector assembly shall consist of a connector tube and a compression ring. For class A stuffing tubes, the connector may be separate components or may be a single piece. Class B and C connectors shall be a single piece.

3.2.3.1 Class A stuffing tubes shall have connector assemblies fabricated of tantalum conforming to the composition specified in table I as determined by spectrographic or other standard methods of chemical analysis.

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Table I - Chemical composition, tantalum connector assembly, class A stuffing tubes.

Oxygen percent, max.	Nitrogen percent max.	Carbon percent, max.	Hydrogen percent, max.	Tantalum
0.015	0.010	0.010	0.002	Remainder

3.2.3.2 Class B stuffing tubes shall have connector assemblies fabricated of silver-plated phosphor-bronze. The silver plating shall be in accordance with type I, class A of QQ-S-365, and shall have an average thickness of 0.0005 inches. The quality assurance tests for solderability (soft solder), and surface roughness, are not required. The phosphor-bronze shall be an alloy conforming to the composition specified in ASTM B-139-58, Alloy B2.

3.2.3.3 Class C stuffing tubes shall have connector assemblies fabricated from zinc-coated phosphor-bronze. The zinc coating shall be in accordance with type I, class 3 of QQ-Z-325, except the quality assurance tests for salt spray are not required and coating thickness may be determined by instrument methods. The phosphor-bronze shall be an alloy conforming to the composition specified in ASTM B-139-58, Alloy B2.

3.2.4 Molded packing (all types). - The molded packing shall have the formulation specified in table II.

Table II - Formulation for molded packing for stuffing tubes.

Component	Part by weight
Geon polyblend 503 (or equivalent)	100
MPC carbon black	5
Stearic acid	1
Dibutyl phthalate	20
Altax or MBTS	1.5
Litharge	10
Crystex (insoluble sulfur)	1.5

In addition, the molded packing shall be identified by a color code to identify the type of connector assembly in accordance with the following, (see figure 1):

Class A	Red
Class B	White
Class C	Green

3.2.5 Gland nut assembly (all types). - The components of the gland nut assembly (gland nut, gland ring, and spring-lock washer) shall be monel metal conforming to the composition specified in QQ-N-281.

3.2.6 Connecting cable. - The connecting cable shall be designation DSS-3 or SHOF-23 conforming to MIL-C-915.

3.3 Manufacture. -

3.3.1 Steel body (all types). - The steel body shall be fabricated by machining a steel bar or forging to the size required by the class required. The threads inside the body shall be 1-12UNF-28 conforming to Handbook H-28.

3.3.2 Teflon bushing (all types). - The teflon bushing shall be fabricated by machining a teflon rod.

3.3.3 Connector assembly. -

3.3.3.1 For class A stuffing tube. - The tantalum connector for the class A stuffing tube shall be fabricated by machining a tantalum bar. The connector may be fabricated from two pieces which are welded together. Welding shall be tungsten inert-gas arc welding.

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3.3.3.2 For class B stuffing tube. - The phosphor-bronze connector assembly for the class B stuffing tube shall be fabricated by machining a phosphor-bronze rod and then silver-plated in accordance with 3.2.3.2.

3.3.3.3 For class C stuffing tube. - The phosphor-bronze connector assembly for the class C stuffing tube shall be fabricated by machining a phosphor-bronze rod and then zinc-coated in accordance with 3.2.3.3.

3.3.4 Molded packing. - The packing shall be molded around the connector assembly and the crimped cable end, holding them in intimate contact. The molded packing shall be bonded to the cable sheath and cured as required for the proper bonding to the particular cable sheath used. The cure for the packing when bonded to thermoplastic jacketed cable (polyvinyl chloride) shall be heat treatment of the bonded packing and cable jacket for 150 minutes at 220°F. The cure for the packing when bonded to thermoset jacketed cable (neoprene) shall be heat treatment of the packing and cable jacket for 90 minutes at 300°F. The flash edges on the packing shall be buffed smooth. Edges projecting greater than 1/32-inch shall not be permitted. A colored dot of Latex paint shall be placed on the cable for identification of the connector type during shipment and installation in accordance with 3.2.3.

3.3.5 Gland nut assembly. - The components of the gland nut assembly shall be fabricated from a monel octagonal or round bar by machining. The threads on the gland nut shall be 1-12UNF-2A conforming to the Handbook H-28.

3.4 Eccentricity of body cavity. - The cavity of the steel body which accepts the hub of the anode or reference electrode shall be positioned axially and concentrically in relation to the outer diameter. Eccentricity in excess of 0.005 inches will not be permitted.

3.5 Marking of stuffing tubes. -

3.5.1 Manufacturer's symbol. - A manufacturer's identifying symbol shall be stamped or engraved on the outer surface of the body. A capital letter corresponding to the revision number of the military specification to which the stuffing box conforms shall also be used in addition to the manufacturer's symbol.

3.6 Workmanship. - The surfaces of all components of the stuffing tube shall be smooth and free of burrs, inclusions, dents, scratches, cuts, and other defects of similar nature consistent with good commercial practice.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. - Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Sampling. -

4.2.1 Lot. - Unless otherwise specified in the contract or order, a lot shall consist of those stuffing tubes offered for delivery at one time or during a fixed period of time, which have been manufactured or processed under substantially the same conditions.

4.2.2 Sampling for examination. - From each lot specified in 4.2.1, stuffing tubes shall be selected at random in accordance with table III for a surface examination and dimensional check.

4.2.3 Sampling for tests. - From each lot specified in 4.2.1, two stuffing tubes shall be selected at random for the tests specified in 4.4.

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Table III - Sampling procedure for examination

Number of stuffing tubes in lot	Number of stuffing tubes selected	Acceptance number	Rejection number
1 to 15	All	---	1
16 to 25	15	1	2
26 to 40	20	1	2
41 to 65	25	2	3
66 to 110	30	2	3
111 to 180	35	3	4
181 and over	50	4	5

4.2.3.1 Resampling. - In cases where one of the two stuffing tubes tested in 4.4 fails to pass the test, at the request of the manufacturer, four (4) additional stuffing tubes may be selected for retest.

4.4 Examination of preparation for delivery. - The preservation, packaging, packing and marking of the stuffing tubes shall be subjected to examination and tests to determine compliance with the requirements of Section 5 of this specification.

5. PREPARATION FOR DELIVERY

5.1 Packaging. - Packaging shall be level A or C, as specified (see 6.2).

5.1.1 Level A. - Stuffing tubes shall be packaged in quantities of one with each package having a net weight of 5-1/2 pounds, unless otherwise specified, and snugly fitted into fiberboard boxes conforming to type II, class 2 of PPP-B-636. The stuffing tubes shall be immobilized within the boxes with solid fiberboard or other suitable separators. The gross weight of the boxes shall not exceed the weight limitations of the fiber box specifications.

5.1.2 Level C. - Preservation and packaging shall be sufficient to afford physical protection against damage during shipment from the supply source to the first receiving activity for immediate use. This level may conform to the suppliers' commercial practice when such meets the requirements of this level.

5.2 Packing. - Packing shall be Level A, B, or C as specified (see 6.2).

5.2.1 Level A. - Stuffing tubes shall be packed in unit pallet loads or in boxes in accordance with the following:

5.2.1.1 Pallets shall conform to NN-P-71 or MIL-P-15011. The stuffing tubes shall be secured to the pallet with tension-tied coated or galvanized steel strapping applied lengthwise and girthwise. Minimum size of the strapping shall be 3/4-inch by 0.033 inch. The gross weight of the pallet load shall not exceed 3,000 pounds.

5.2.1.2 The boxes shall conform to PPP-B-601 (overseas type) or class 2 of PPP-B-621. The gross weight of the box shall not exceed approximately 500 pounds. Box closure and strapping shall be in accordance with the applicable box specification or appendix thereto. The boxes shall be modified by the addition of skids, for gross weights exceeding 200 pounds, in accordance with the applicable box specification.

5.2.1.3 The stuffing tubes shall be cushioned, blocked and braced in accordance with JAN-P-100.

5.2.2 Level B. - Stuffing tubes shall be packed in unit pallet loads or in boxes in accordance with the following:

5.2.2.1 Pallets shall be as specified in 5.2.1.1.

5.2.2.2 The boxes shall conform to class 1 of PPP-B-621, or PPP-B-601 (domestic type). The gross weight of the box shall not exceed approximately 500 pounds. Box closure shall be in accordance with the applicable box specification or appendix thereto. The boxes shall be modified by the addition of skids, for gross weights exceeding 200 pounds, in accordance with the applicable box specification.

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5.2.2.3 The stuffing tubes shall be cushioned, blocked and braced so as to prevent damage during shipment.

5.2.3 Level C. - Stuffing tubes shall be packed in containers, at the lowest rates, in a manner which will insure acceptance by common carrier and will afford protection against physical or mechanical damage during direct shipment from the supply source to the first shipment from the supply source to the first receiving activity for immediate use. This level in general shall conform to the Uniform Freight Classification Rules and Regulations or other carrier regulations as applicable to the mode of transportation and may be the supplier's commercial practice when such meets the requirements of this level.

5.3 Marking. - In addition to any special marking required by the contract or order, marking of the packages and shipping containers shall be in accordance with MIL-STD-129.

6. NOTES

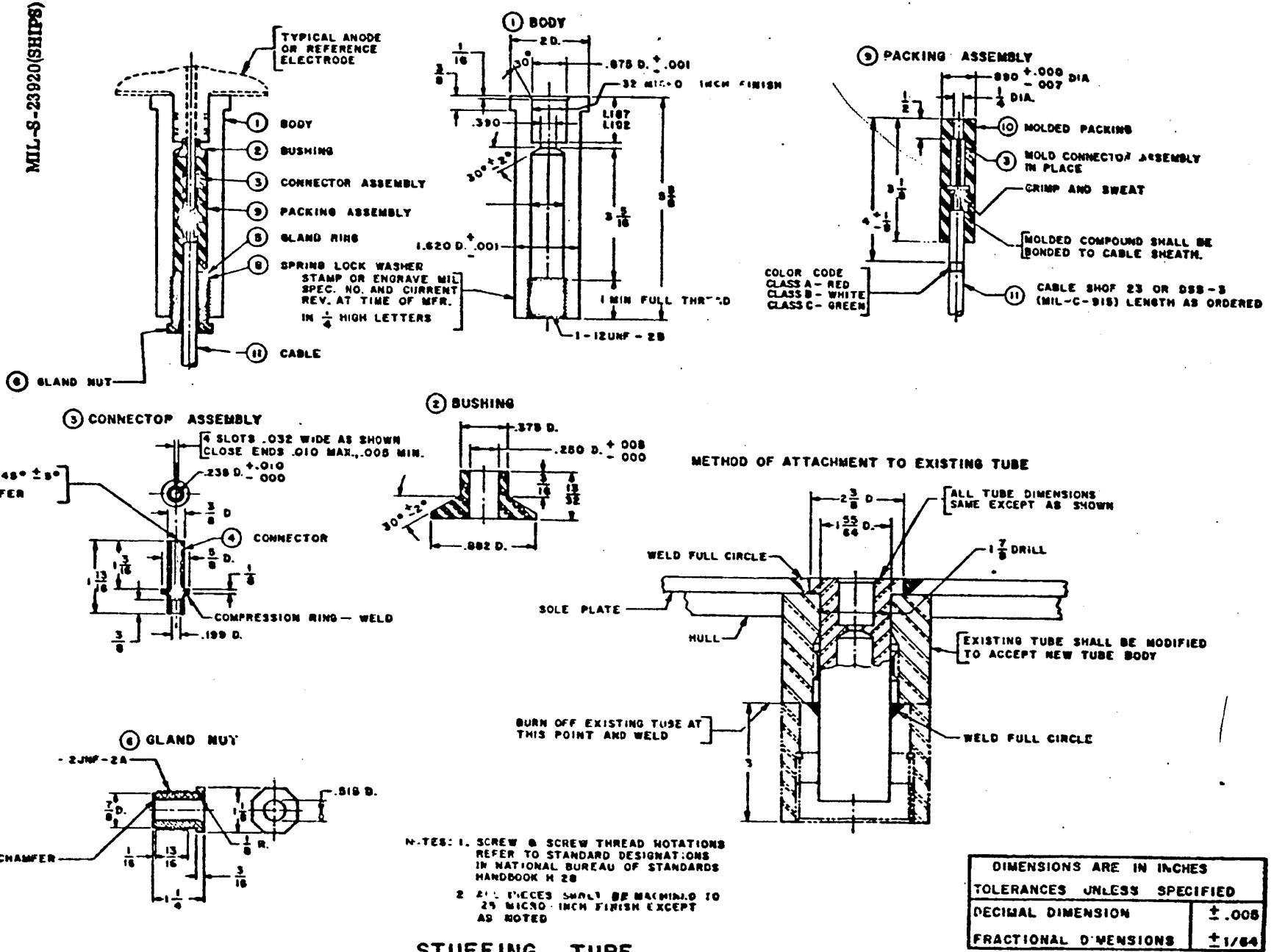
6.1 Intended use. - The stuffing tubes are primarily intended for use in corrosion prevention in sea water of ship hulls, steel and aluminum equipment and structures, seachests, sonar domes, and similar marine structures when used in conjunction with circular platinum-palladium anodes, silver-silver chloride reference electrodes, or zinc reference electrodes.

6.2 Ordering data. - Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Type and class of stuffing tube and alternate body material if required (see 1.2).
(c) Type and length of cable attached to the stuffing tube.
(d) Level of preservations, packaging, and packing required (see 5.1 and 5.2).
(e) Special marking if required (see 5.3).
(f) Number of stuffing tubes per package (see 5.1).

Preparing activity:
Navy - Ships
(Project 2040-N051Sh)

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NOTES: 1. SCREW & SCREW THREAD NOTATIONS REFER TO STANDARD DESIGNATIONS IN NATIONAL BUREAU OF STANDARDS HANDBOOK H 28

2. ALL PIECES SHALL BE MACHINED TO 25 MICRO-INCH FINISH EXCEPT AS NOTED

DIMENSIONS ARE IN INCHES	
TOLERANCES UNLESS SPECIFIED	
DECIMAL DIMENSION	±.008
FRACTIONAL DIMENSIONS	±1/64

STUFFING TUBE
FIGURE 1

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SPECIFICATION ANALYSIS SHEET

Form Approved Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity.

SPECIFICATION

ORGANIZATION	CITY AND STATE

CONTRACT NO.	QUANTITY OF ITEMS PROCURED	DOLLAR AMOUNT
		$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*	DATE

DD FORM 1 APR 63 1426 REPLACES NAVSHIPS FORM 4063, WHICH IS OBSOLETE

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